

Solution to the Daily Limit

Based on the problem provided in the file [dailyintegral-limit-limit-1x1-2026-08-19.jpg](#), we are tasked with evaluating the following limit:

$$L = \lim_{x \rightarrow \infty} \left[\left(\ln \left(1 + \frac{1}{x} \right) \right)^{-\left(\ln \left(1 + \frac{1}{x} \right) \right)^{\frac{1}{x}}} - x + \left(\ln \left(\ln \left(1 + \frac{1}{x} \right) \right) \right)^2 \right]$$

1. Variable Substitution

To make the expression easier to work with, we can introduce a new variable. Let $t = \ln \left(1 + \frac{1}{x} \right)$. As $x \rightarrow \infty$, $\frac{1}{x} \rightarrow 0^+$, which means $t \rightarrow \ln(1) = 0^+$.

From this substitution, we can also express x in terms of t :

$$\begin{aligned} e^t &= 1 + \frac{1}{x} \\ \frac{1}{x} &= e^t - 1 \\ x &= \frac{1}{e^t - 1} \end{aligned}$$

Substituting t into the original limit L , we get:

$$L = \lim_{t \rightarrow 0^+} \left[t^{-t^{(e^t-1)}} - \frac{1}{e^t - 1} + (\ln t)^2 \right]$$

2. Asymptotic Expansions

Let's break down the expression into manageable parts and find their asymptotic expansions (Taylor series) as $t \rightarrow 0^+$. Let the first term be $A = t^{-t^{(e^t-1)}}$. We can rewrite A using the base e :

$$A = \exp \left(-t^{(e^t-1)} \ln t \right) = \exp \left(-e^{(e^t-1) \ln t} \ln t \right)$$

First, expand the exponent $(e^t - 1) \ln t$:

$$\begin{aligned} e^t - 1 &= t + \frac{t^2}{2} + \mathcal{O}(t^3) \\ (e^t - 1) \ln t &= t \ln t + \frac{t^2}{2} \ln t + \mathcal{O}(t^3 \ln t) \end{aligned}$$

Next, expand the exponential $e^{(e^t-1) \ln t}$ using $e^u = 1 + u + \frac{u^2}{2} + \dots$:

$$e^{(e^t-1) \ln t} = 1 + (t \ln t) + \frac{1}{2}(t \ln t)^2 + \mathcal{O}(t^2 \ln t)$$

Now substitute this back into the exponent of A , multiplying by $-\ln t$:

$$\begin{aligned} -e^{(e^t-1) \ln t} \ln t &= - \left(1 + t \ln t + \frac{1}{2} t^2 \ln^2 t + \dots \right) \ln t \\ &= -\ln t - t \ln^2 t - \frac{1}{2} t^2 \ln^3 t + \mathcal{O}(t^3 \ln^4 t) \end{aligned}$$

Exponentiating this result to find A :

$$\begin{aligned}
A &= \exp(-\ln t - t \ln^2 t + \mathcal{O}(t^2 \ln^3 t)) \\
&= \exp(-\ln t) \cdot \exp(-t \ln^2 t + \mathcal{O}(t^2 \ln^3 t)) \\
&= \frac{1}{t} \left(1 - t \ln^2 t + \frac{1}{2}(-t \ln^2 t)^2 + \dots \right) \\
&= \frac{1}{t} - \ln^2 t + \mathcal{O}(t \ln^4 t)
\end{aligned}$$

Next, let's expand the second term of the limit limit, $\frac{1}{e^t - 1}$:

$$\begin{aligned}
\frac{1}{e^t - 1} &= \frac{1}{t + \frac{t^2}{2} + \frac{t^3}{6} + \mathcal{O}(t^4)} \\
&= \frac{1}{t} \left(1 + \frac{t}{2} + \frac{t^2}{6} + \mathcal{O}(t^3) \right)^{-1} \\
&= \frac{1}{t} \left(1 - \frac{t}{2} + \mathcal{O}(t^2) \right) \\
&= \frac{1}{t} - \frac{1}{2} + \mathcal{O}(t)
\end{aligned}$$

3. Evaluating the Limit

Now, substitute the asymptotic expansions back into the full expression for L :

$$L = \lim_{t \rightarrow 0^+} \left[\left(\frac{1}{t} - \ln^2 t + \mathcal{O}(t \ln^4 t) \right) - \left(\frac{1}{t} - \frac{1}{2} + \mathcal{O}(t) \right) + (\ln t)^2 \right]$$

Group the terms together:

$$L = \lim_{t \rightarrow 0^+} \left[\left(\frac{1}{t} - \frac{1}{t} \right) + (-\ln^2 t + \ln^2 t) + \frac{1}{2} + \mathcal{O}(t \ln^4 t) \right]$$

The divergent terms $\frac{1}{t}$ and $\ln^2 t$ perfectly cancel each other out:

$$L = \lim_{t \rightarrow 0^+} \left[\frac{1}{2} + \mathcal{O}(t \ln^4 t) \right]$$

Since $\lim_{t \rightarrow 0^+} t \ln^4 t = 0$, the remaining limit is straightforward.

Conclusion

Therefore, the evaluated limit is:

$$\lim_{x \rightarrow \infty} \left[\left(\ln \left(1 + \frac{1}{x} \right) \right)^{-\left(\ln \left(1 + \frac{1}{x} \right) \right)^{\frac{1}{x}}} - x + \left(\ln \left(\ln \left(1 + \frac{1}{x} \right) \right) \right)^2 \right] = \frac{1}{2}$$